

Project Title: SE Chain Simulator – Social Engineering Security Training Simulator

1. Project Overview

The **SE Chain Simulator** is a Python-based cybersecurity training project developed as the final project of the Srock Cybersecurity Internship.

The project consolidates the cybersecurity and social-engineering concepts practiced during Days 1–14 into a single **menu-driven simulation application**.

The simulator provides five security-training modules:

1. **OSINT** – Passive OSINT on a domain
2. **Profile** – Target profile building from public information
3. **Phish** – Phishing URL indicator analysis
4. **Template** – Spear-phishing awareness template generation
5. **IR** – Incident-response workflow

The project is designed for **educational and authorized cybersecurity training purposes**.

2. Problem Statement

Social engineering attacks can involve several stages, including information gathering, target profiling, phishing, and incident response.

During the internship, these concepts were studied separately. The final project combines them into one simple application so that the overall security workflow can be demonstrated through a single interface.

3. Project Objectives

The main objectives are:

- To consolidate the concepts learned during the internship.
- To develop a Python-based cybersecurity simulation.
- To provide a simple menu-driven interface.
- To demonstrate different social-engineering security concepts.
- To understand phishing indicators and security awareness.
- To demonstrate incident-response concepts.
- To practice Python programming and project organization.
- To maintain the project in a GitHub repository.

4. System Modules

Module 1 — OSINT

OSINT stands for Open-Source Intelligence.

This module represents the passive collection of publicly available information about a domain.

Purpose:

- Understand passive reconnaissance.
- Identify publicly available information.
- Demonstrate how information can be collected without directly attacking a system.

Module 2 — Profile

The Profile module represents building a security-awareness target profile using publicly available information.

Purpose:

- Understand information exposure.
- Identify information that could potentially be useful in social engineering.
- Demonstrate the importance of limiting unnecessary public information.

Module 3 — Phish

The Phish module represents phishing URL analysis.

It focuses on identifying suspicious characteristics in URLs.

Examples of indicators include:

- Suspicious domains
- Unusual URL structure
- Account/login-related keywords
- Multiple subdomains
- IP-address-based URLs

Purpose:

To improve phishing detection and security awareness.

Module 4 — Template

The Template module represents a **spear-phishing awareness scenario**.

It is intended for training and demonstrates common phishing characteristics such as:

- Urgency
- Unexpected account-security requests
- Verification requests
- Suspicious links

- Sender verification

Important: The project is simulation-only and is not intended to send real phishing messages.

Module 5 — IR

IR means Incident Response.

This module represents the defensive response after a suspected security incident.

Possible response concepts include:

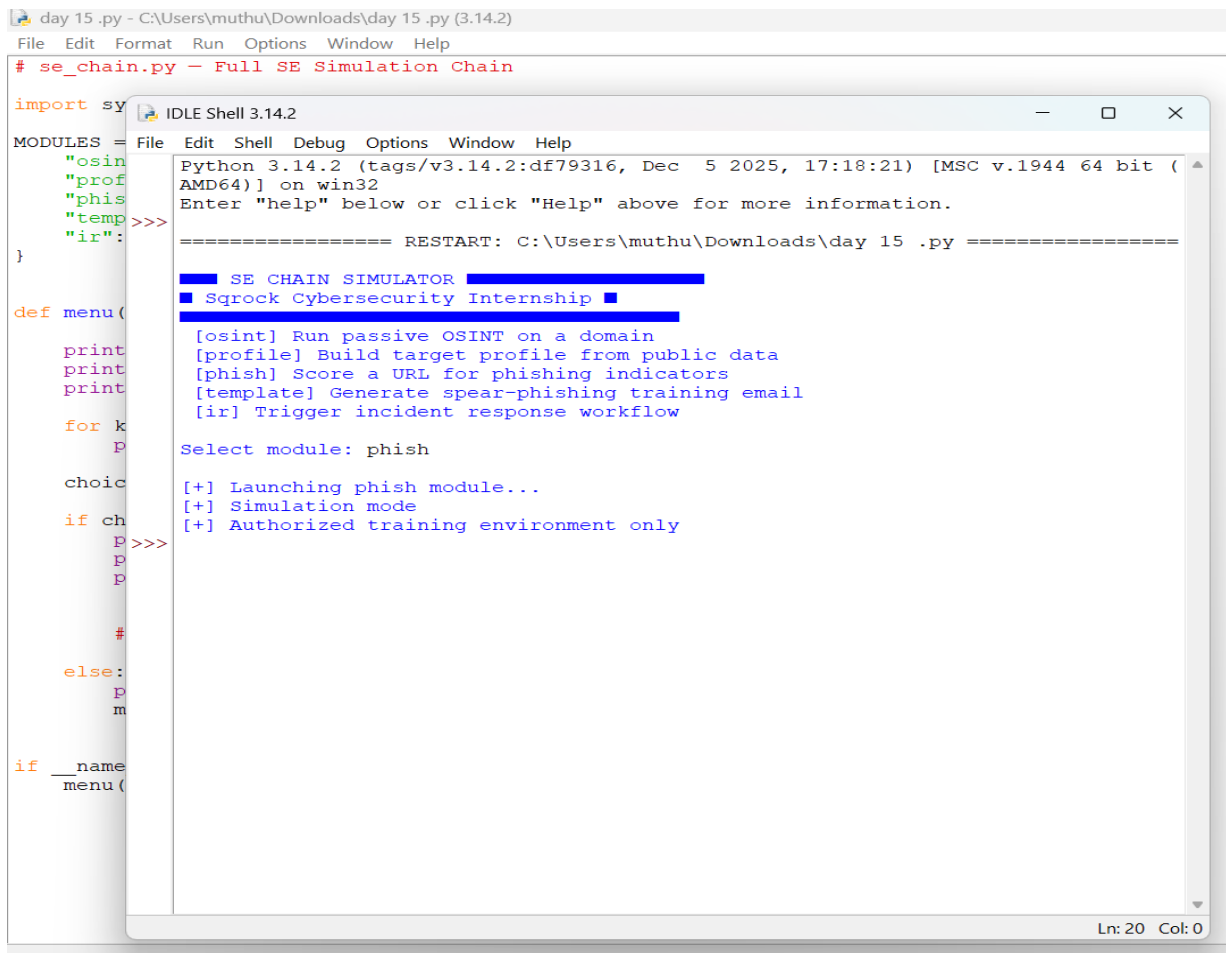
- Account containment
- Session revocation
- Security-team notification
- Log preservation
- Investigation
- Recovery

5. Technology Used

The project uses:

- Python 3
- Python functions
- Dictionaries
- Loops
- Conditional statements
- User input
- String processing
- Git and Git hub

6. Project Output



```
day 15.py - C:\Users\muthu\Downloads\day 15.py (3.14.2)
File Edit Format Run Options Window Help
# se_chain.py - Full SE Simulation Chain

import sys
MODULES = {
    "osint": "Run passive OSINT on a domain",
    "profile": "Build target profile from public data",
    "phish": "Score a URL for phishing indicators",
    "template": "Generate spear-phishing training email",
    "ir": "Trigger incident response workflow"
}

def menu():
    print("\n===== SE CHAIN SIMULATOR =====")
    print("■ Sgrock Cybersecurity Internship ■")
    print("\n[osint] Run passive OSINT on a domain")
    print("[profile] Build target profile from public data")
    print("[phish] Score a URL for phishing indicators")
    print("[template] Generate spear-phishing training email")
    print("[ir] Trigger incident response workflow")
    select = input("Select module: ")
    if select == "phish":
        print("[+] Launching phish module...")
        print("[+] Simulation mode")
        print("[+] Authorized training environment only")
    else:
        print("Invalid module selected.")

if __name__ == "__main__":
    menu()
```

7. Learning Outcomes

Through this final project, I learned:

- Basics of OSINT and passive reconnaissance.
- Phishing URL detection concepts.
- Social-engineering attack indicators.
- Security-awareness practices.
- Target-profile analysis.
- Password-security and rate-limiting concepts.
- Honeypot and monitoring concepts.
- SIEM log analysis.

- Incident-response concepts.
- Python programming for cybersecurity.
- GitHub project management and documentation.

8. Ethical Considerations

This project is developed strictly for **educational cybersecurity training**.

The simulator should only be used in:

- Authorized laboratory environments
- Local systems
- Training environments
- Systems where explicit permission has been provided

It should not be used to target real people, accounts, networks, or websites without authorization.

9. Conclusion

The **SE Chain Simulator** is the final project of my Sgrock Cybersecurity Internship Phase 1.

It consolidates the concepts learned throughout the internship into a single Python-based security-training application. The project demonstrates how reconnaissance, profiling, phishing awareness, security training, and incident response can be represented as connected stages of a cybersecurity workflow.

The project helped me improve both my **Python programming skills** and my understanding of **social engineering, phishing detection, cybersecurity awareness, monitoring, and incident response**.